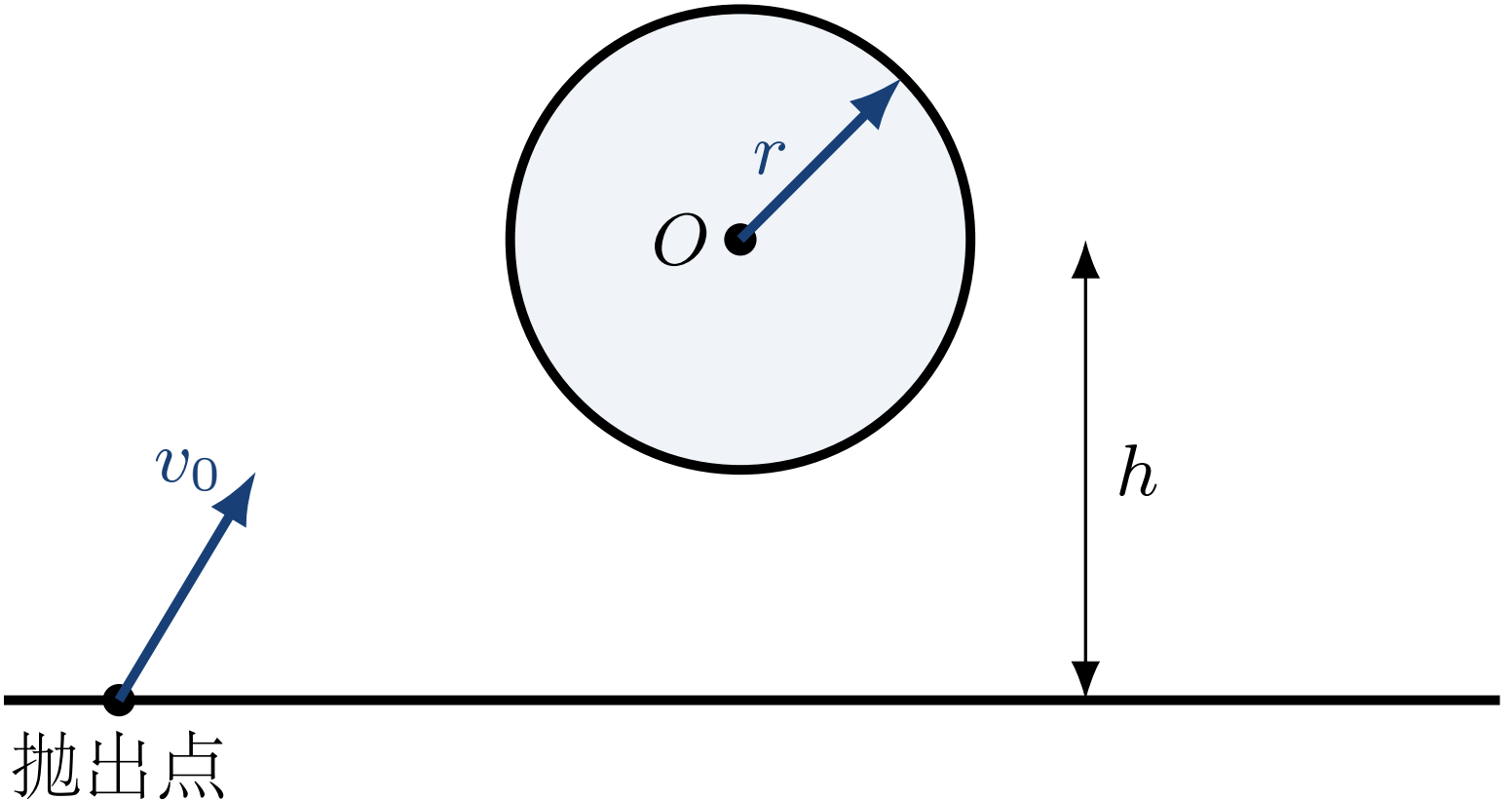
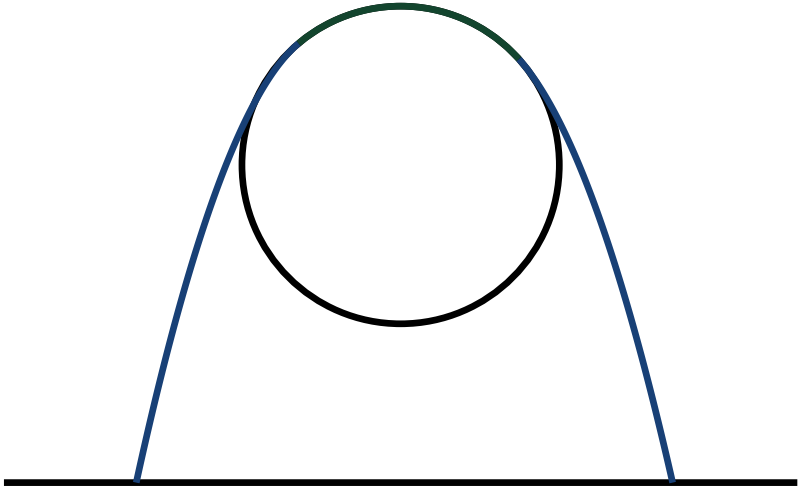


圆柱截面，抛出位置与角度均可选择

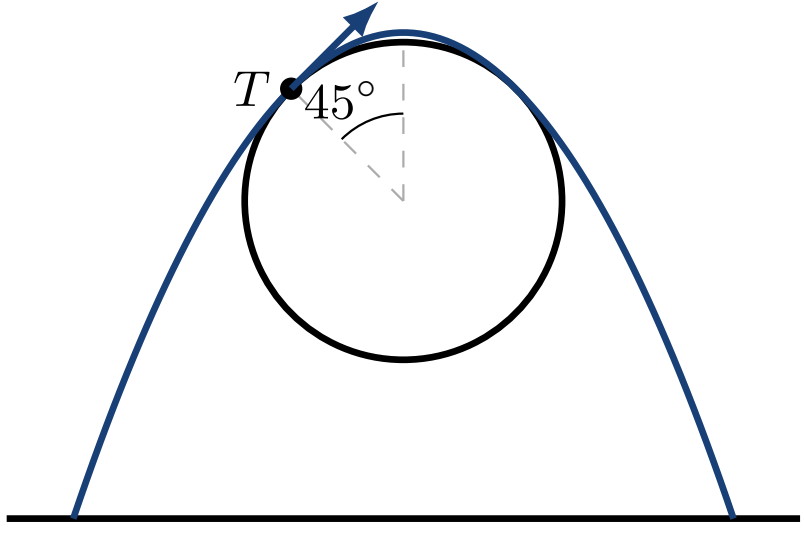
允许接触：势垒 $h + r$



临界速度仅为下确界

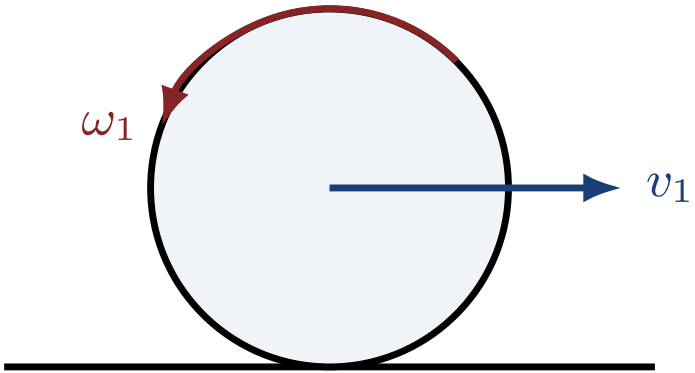
$\cos\varphi_{\text{detach}} = 2/3$

禁止接触：极限抛物线



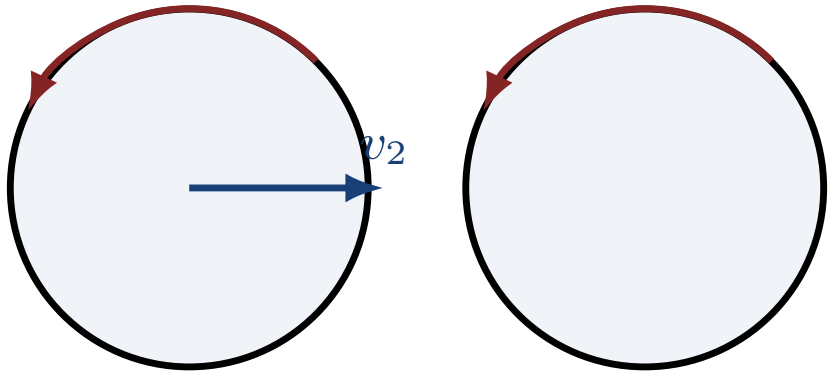
两侧均在 $45°$ 处相切

实际轨迹须留正间隙

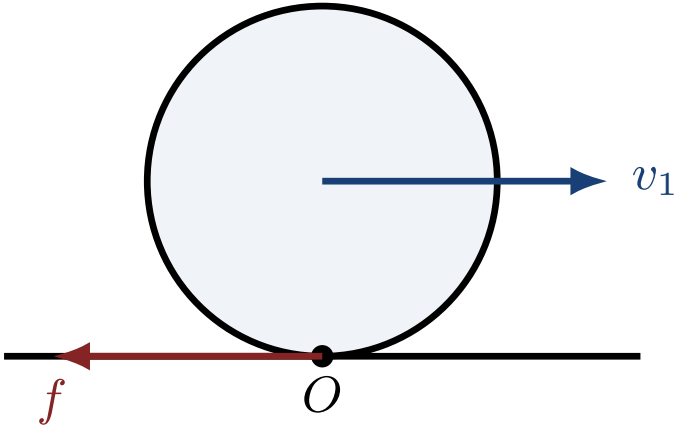


侧视；足够大逆旋可使球反向

俯视：两球自转同向，均为 ω_2

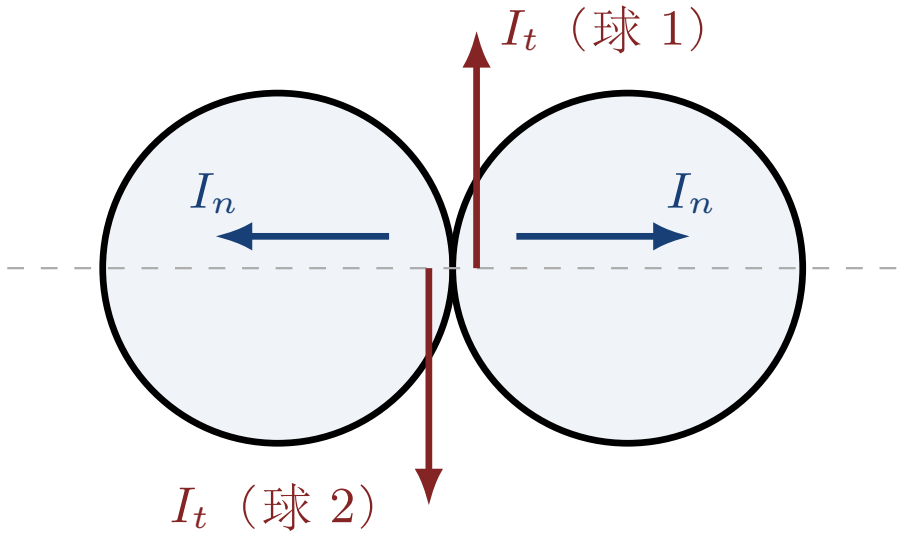


沿球心连线正碰；接触处有切向滑动



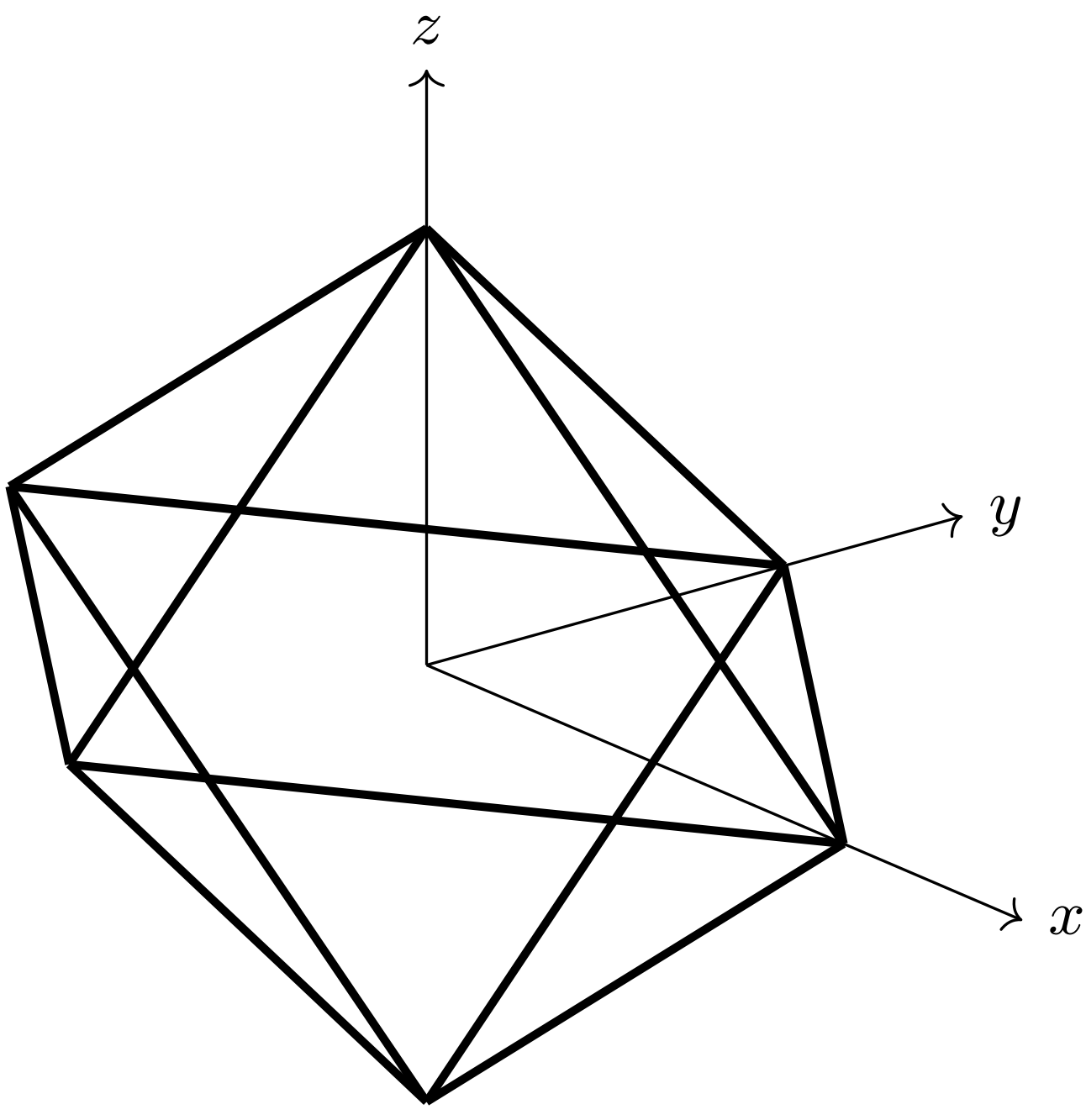
$$L_O = mv_1R - J\omega_1$$

反向滚回须 $J\omega_1 > mv_1R$

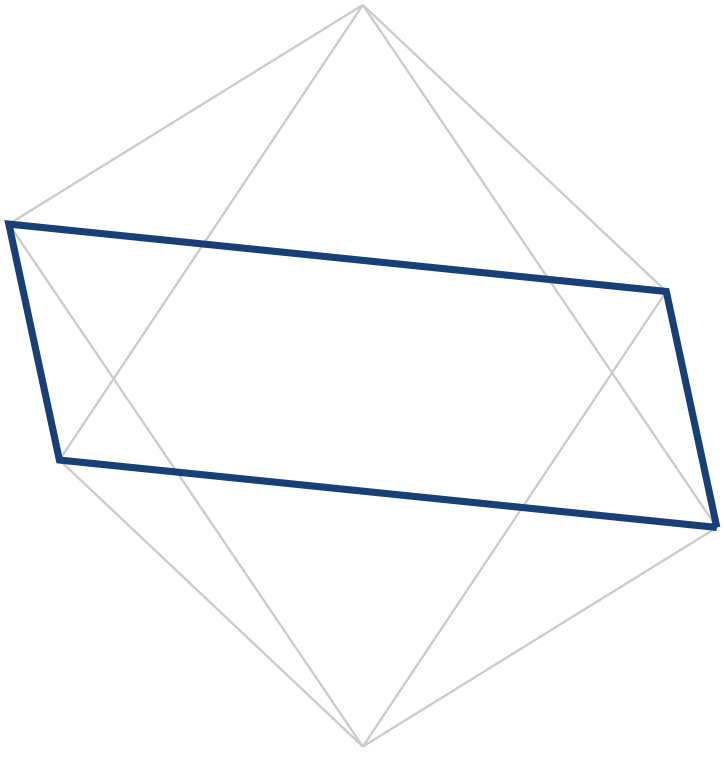


$$I_n = mv_2;\ \ I_t \le \mu I_n$$

$$I_t = mJR\omega_2/(J + mR^2)$$

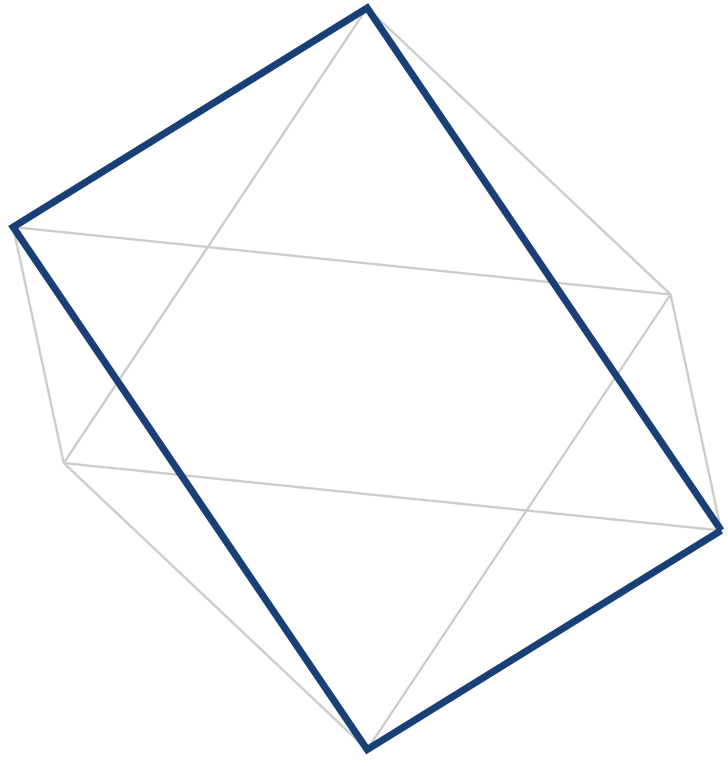


六个顶点位于三个坐标轴上，每条棱长度 l、电阻 r



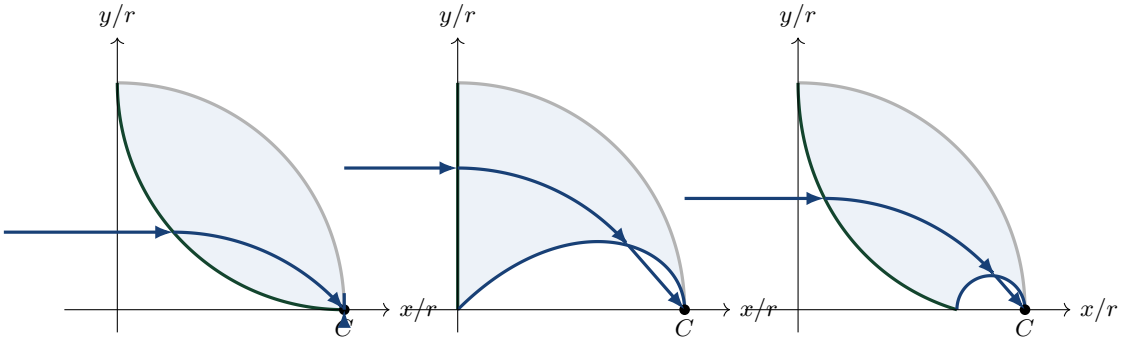
z 分量：$I_z = bl^2/(4\sqrt{3}r)$

回路面积均为 l^2，电阻均为 $4r$



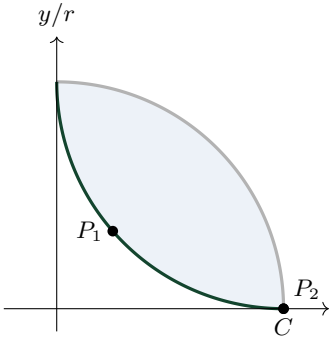
y 分量：$I_y = \sqrt{2}bl^2/(4\sqrt{3}r)$

回路面积均为 l^2，电阻均为 $4r$

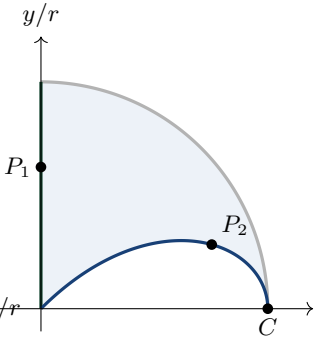


第（1）问　　第（2）问　　第（3）问

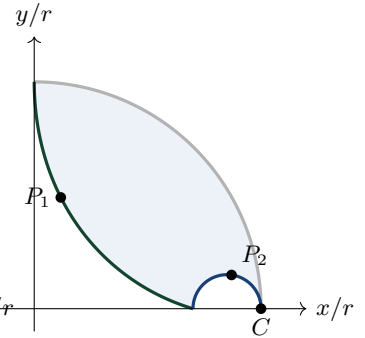
示例 $\ell = .3r\cos\theta$



第（1）问

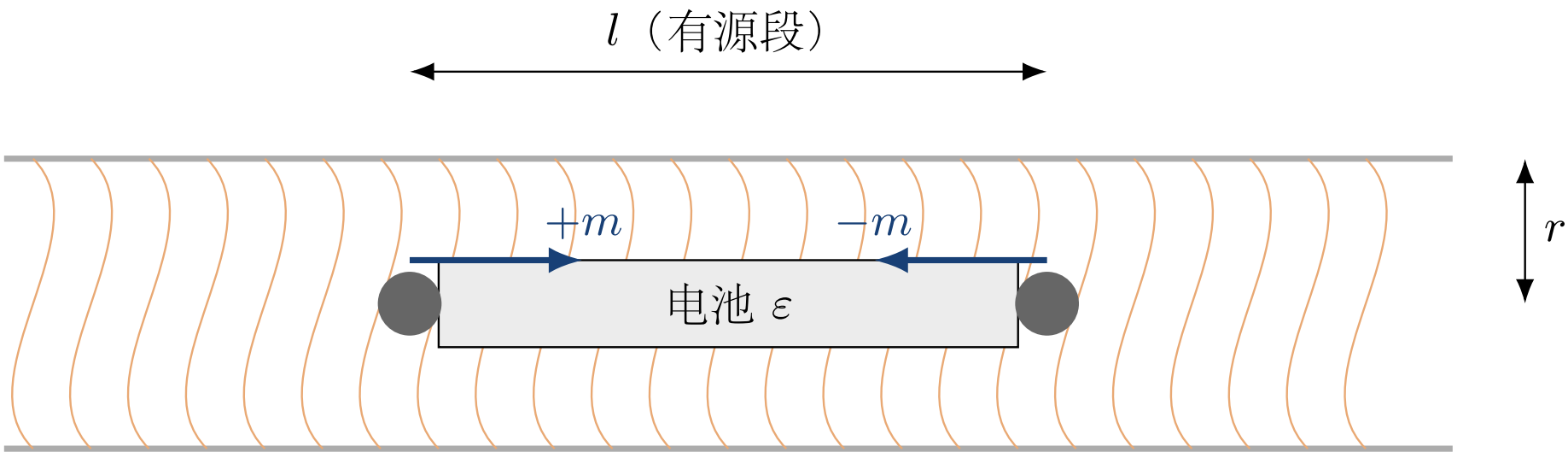


第（2）问



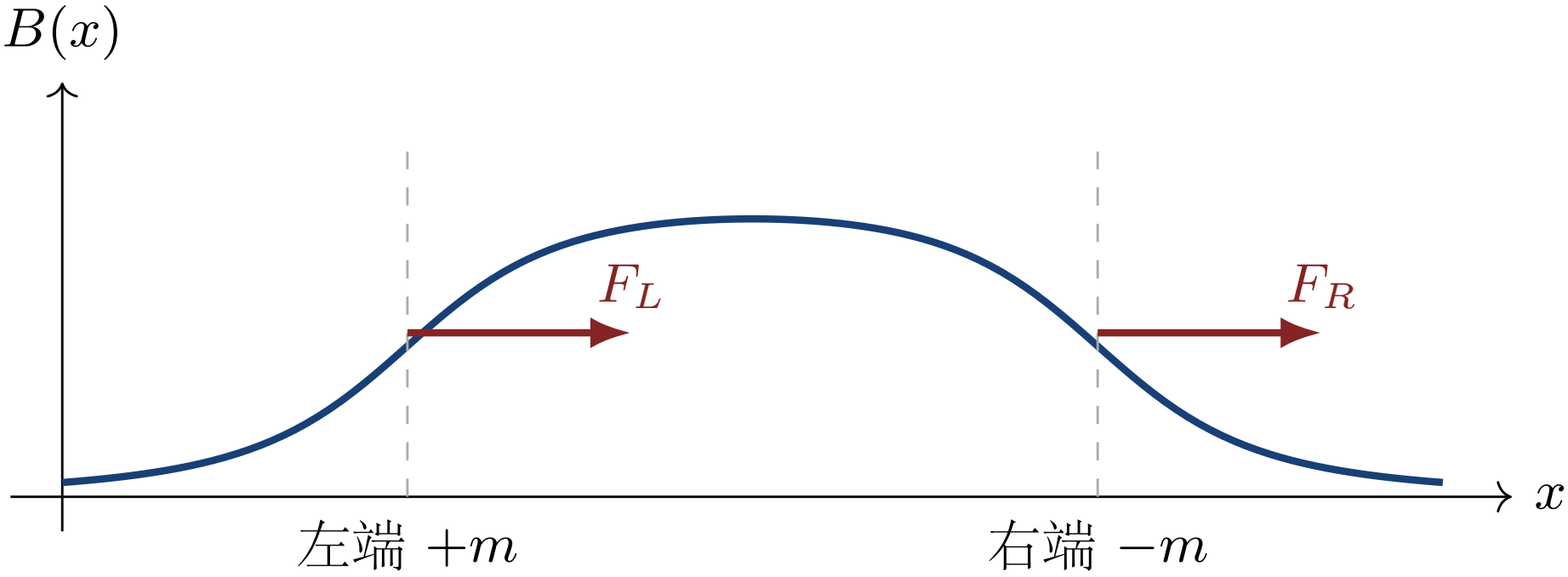
第（3）问

示例 $\ell = .3r\cos\theta$



线圈其余部分沿很大的闭合环返回

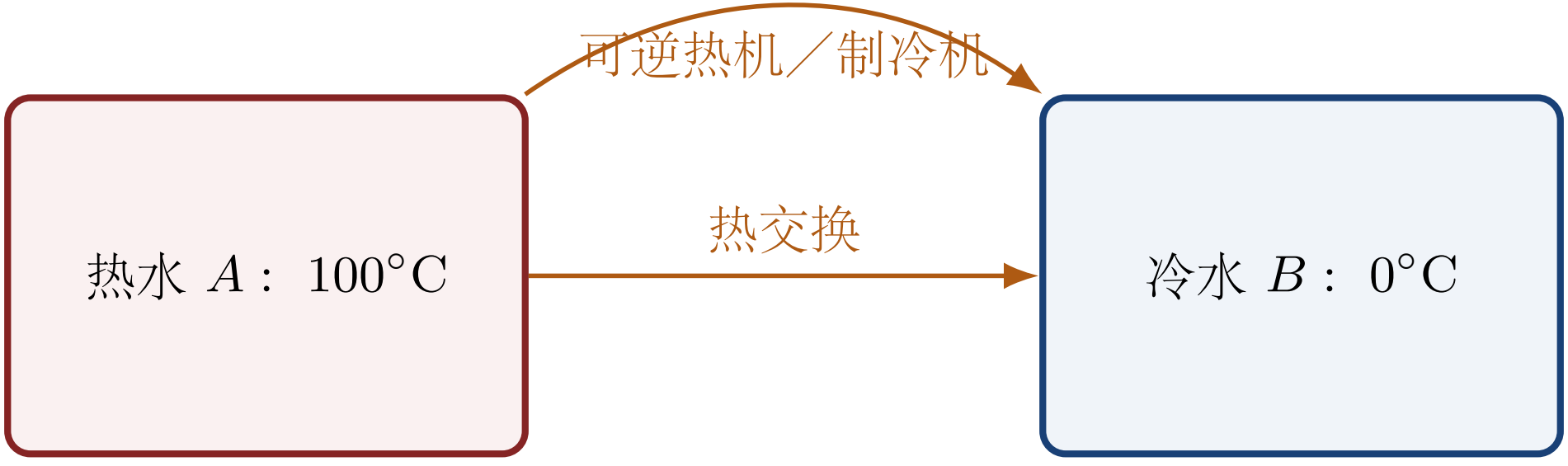
仅画局部，磁体接触处接通长度 l 的线圈段



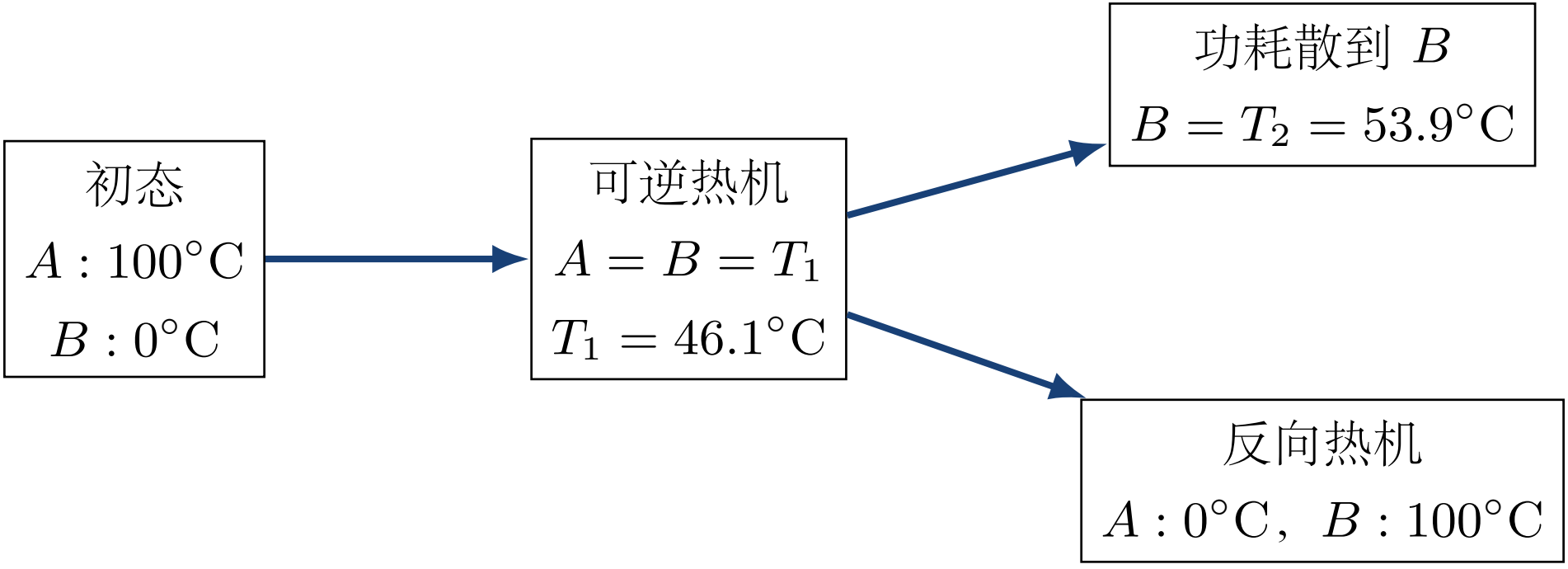
两端力同向：$F = \mathcal{A}I$

$$\varepsilon = IR + \mathcal{A}v,\ \ kv = \mathcal{A}I$$

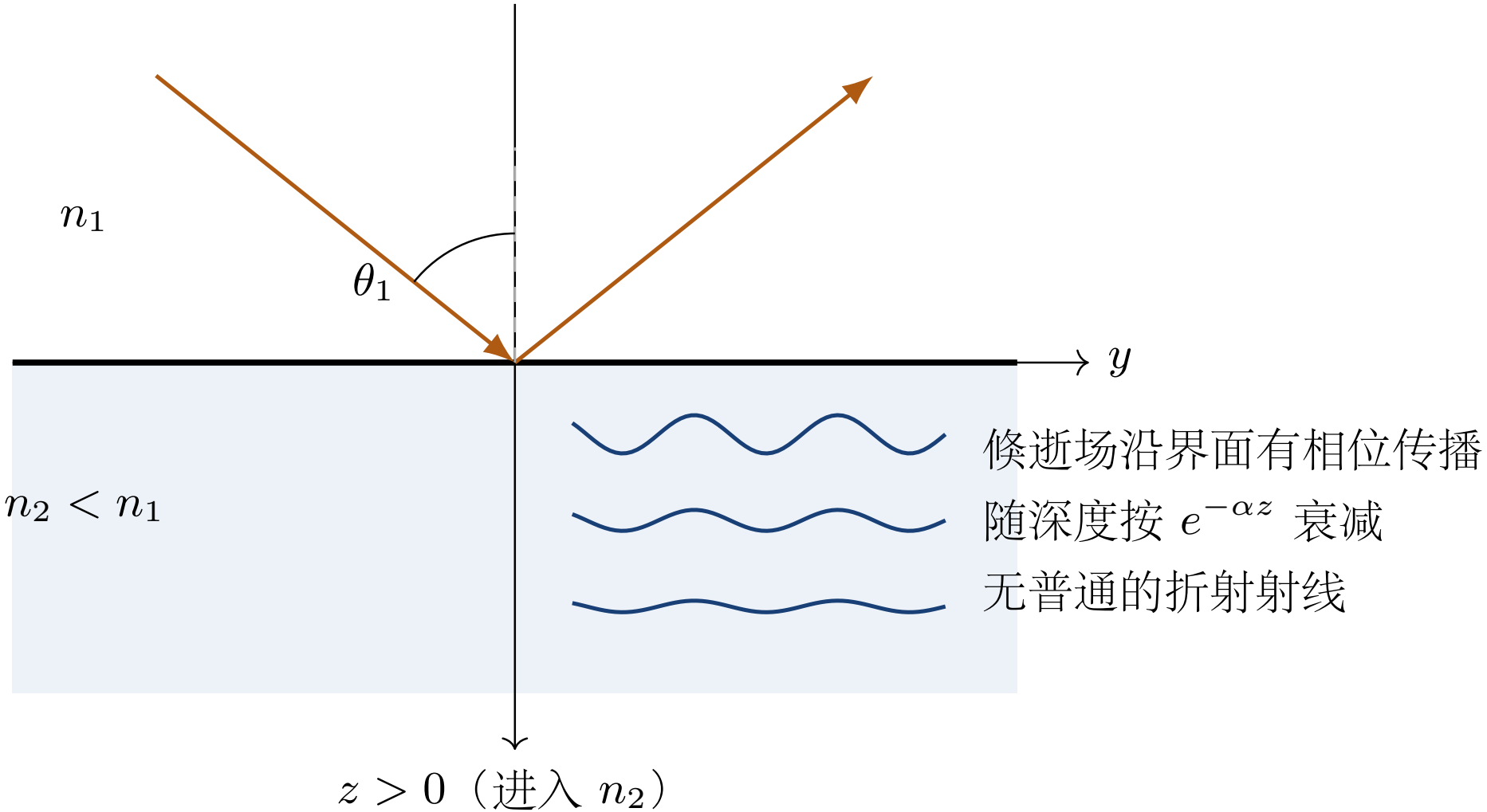
$$v = \mathcal{A}\varepsilon/(kR + \mathcal{A}^2)$$

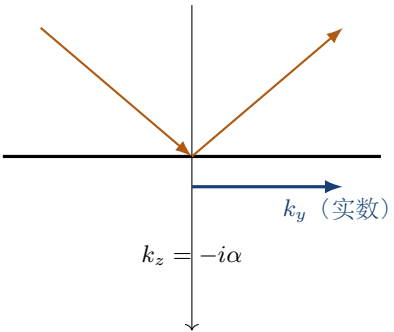


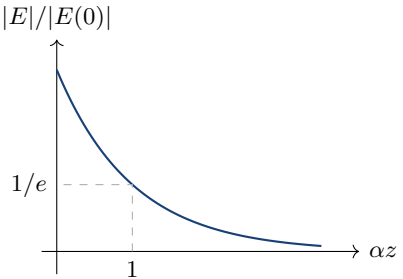
两杯等质量、等比热的水



熵式使用 K：$T_1=\sqrt{373.15\times273.15}\ \mathrm{K}$

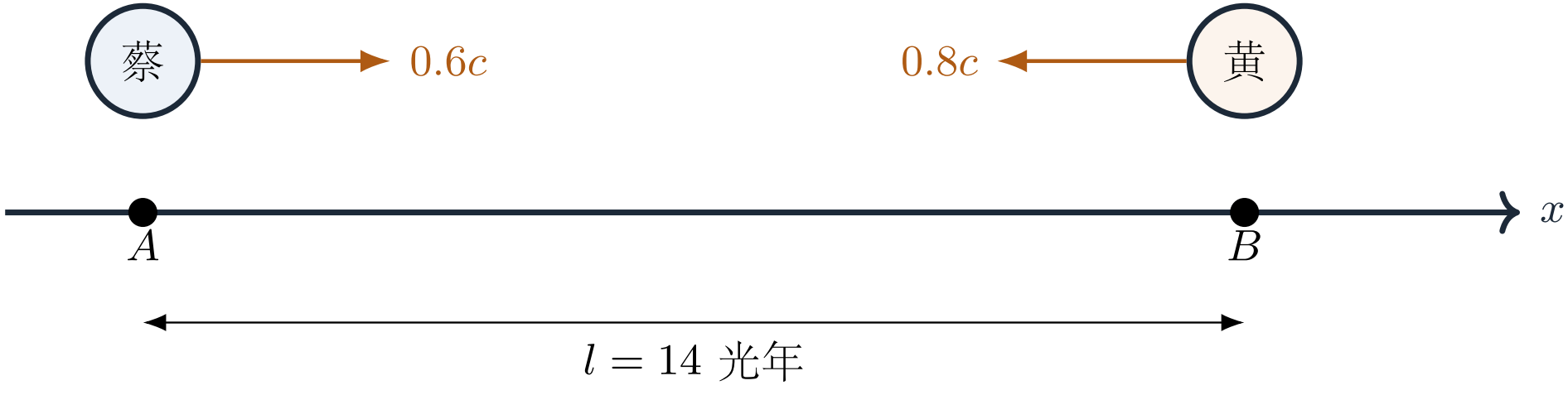

n_1
θ_1
y
$n_2 < n_1$
倏逝场沿界面有相位传播
随深度按 $e^{-\alpha z}$ 衰减
无普通的折射射线
$z > 0$（进入 n_2）



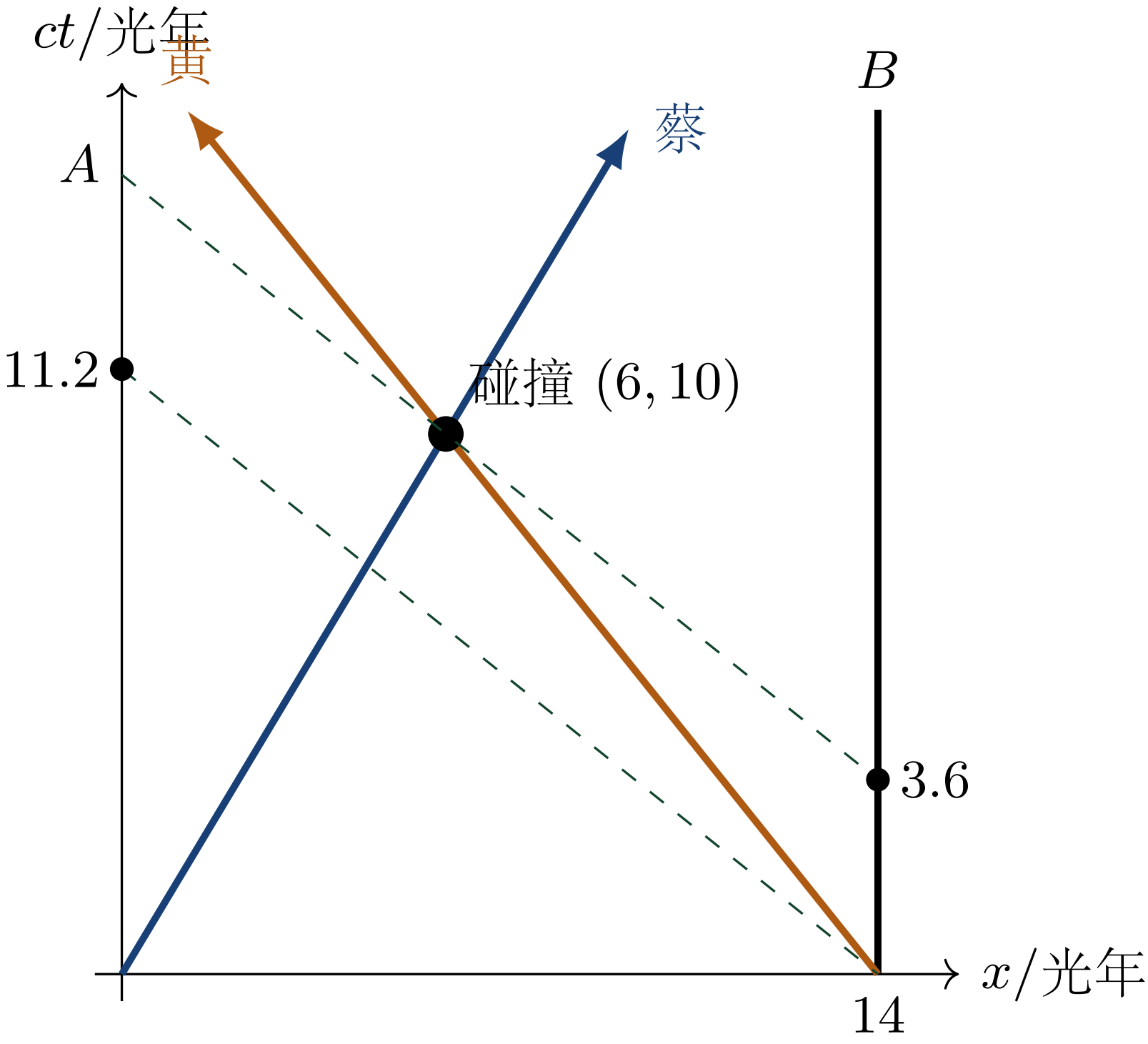


振幅深度 $1/\alpha$

能量密度深度 $1/(2\alpha)$



唐静止参考系中的两村庄与相向运动



两条黄系同时线均满足 $ct + .8x =$ 常量

交点严格位于两条人物世界线上